In Silico Evidence Curation and Pathogenicity Reclassification of MYOD1 Variants of Uncertain Significance in Rhabdomyosarcoma

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Abstract

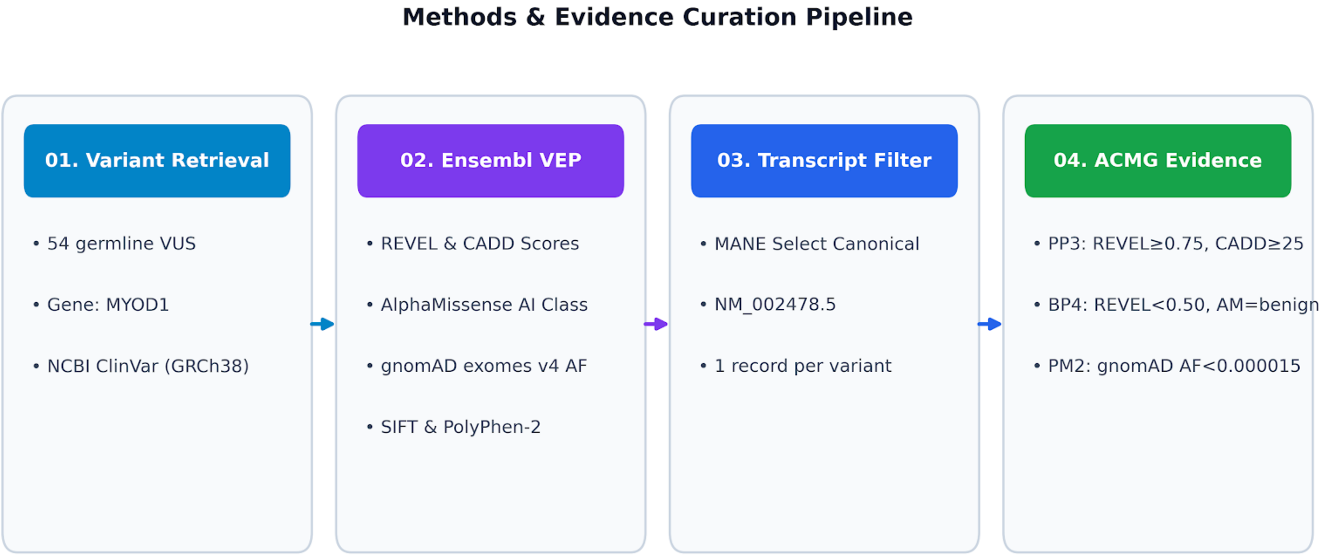
In pediatric oncogenes, missense Variants of Uncertain Significance (VUS) pose a major challenge in clinical genetics and cancer predisposition screening. *MYOD* encodes a master basic helix-loop-helix (bHLH) transcription factor vital for myogenic differentiation, and specific alterations are key drivers in embryonal and sclerosing/spindle cell Rhabdomyosarcoma (RMS). For this study, 54 germline missense VUS entries for MYOD1 were extracted from NCBI ClinVar and batch annotated using ENSEMBL VEP. Cross-referencing allele frequencies against gnomAD exomes and in-silico pathogenicity meta predictors (REVEL, CADD, PHRED and Apha Missense) were also integrated according to the ACMG/AMP variant interpretation framework. 57.4% variants demonstrated benign-leaning predictions, whereas 13% exhibited strong, concordant in-silico pathogenicity. 29.6% remained ambiguous or exhibited intermediate scores. This systematic in-silico evidence curation effectively stratifies *MYOD1* VUS entries, prioritising specific deleterious candidates for downstream wet-lab validation in Rhabdomyosarcoma models.

1. Introduction

Rhabdomyosarcoma (RMS) is a pediatric soft tissue sarcoma with four major subtypes, as recognised by WHO: embryonal RMS (ERMS), alveolar RMS (ARMS), pleomorphic RMS (PRMS), and sclerosing/spindle RMS. While predominantly ERMS and ARMS are observed in most patients, ERMS is the most common subtype, followed by ARMS. ARMS is predominantly driven by chimeric fusion genes (*PAX3/7-FOXO1*). ERMS frequently harbour single-nucleotide variants in key myogenic regulators and oncogenic pathways [1].

Myogenic Differentiation 1 (*MYOD1*) encodes a master basic helix-loop-helix (bHLH) transcription factor that orchestrates muscle cell lineage commitment and differentiation by binding to canonical E-box consensus sequences (5'-CANNTG-3'). The myogenic differentiation 1 gene (*MYOD1*) p.L122R mutation act as a blockage for cell differentiation [2]. However, many of the missense variants across the *MYOD1* gene are currently classified in ClinVar as Variants of Uncertain Significance (VUS). When clinical geneticists encounter a VUS, it cannot be used to guide diagnosis or risk assessment [3]. Reducing this burden of uncertainty through systematic in-silico curation and evidence synthesis is an essential first step toward candidate prioritization and clinical utility [4].

2. Methods



1. **Variant Retrieval:** All 54 germline missense VUS entries listed for *MYOD1* were downloaded in tsv format from NCBI ClinVar (Assembly GRCh38) [5], [6].

2. **Batch Annotation via Ensembl VEP:** Cleaned HGVS transcript strings (NM_002478.5) were processed through Ensembl Variant Effect Predictor (VEP) to retrieve [6]:

   - Ensemble pathogenicity scores via REVEL and CADD PHRED
   - Deep-learning predictions via AlphaMissense (likely_pathogenic, likely_benign, or ambiguous)
   - Population allele frequencies via gnomAD exomes v4
   - Standard reference predictors- SIFT and PolyPhen-2

3. **Canonical Transcript Filtering:** Output records were filtered for the MANE Select canonical transcript NM_002478.5 to ensure one consistent annotation per variant.

4. **ACMG/AMP Evidence Rules Applied:**

   - **PP3 (Supporting Pathogenic):** Applied if REVEL ≥ 0.75, CADD PHRED ≥ 25.0, AND AlphaMissense = *likely_pathogenic*
   - **BP4 (Supporting Benign):** Applied if REVEL < 0.50 AND AlphaMissense = *likely_benign*
   - **PM2 (Supporting Pathogenic):** Applied if the variant was absent or extremely rare in gnomAD (AF<0.000015) [6], [7].

3. Results & Visualizations

3.1 High-Priority Pathogenic Candidates

Systematic batch annotation and filtering of all 54 germline missense VUS in *MYOD1* revealed a distinct, bimodal distribution of computational pathogenicity signals. By evaluating ensemble machine-learning scores (REVEL and CADD PHRED), deep-learning structural predictions (AlphaMissense), and population allele frequencies (gnomAD exomes v4), 31 out of 54 variants (57.4%) met the criteria for benign-leaning classification (BP4, PM2). These variants exhibited REVEL scores under 0.50, CADD scores below 25.0, and unanimous *likely_benign* classifications by AlphaMissense, despite being rare or absent in population databases. Whereas 7 variants met all the criteria for strong in-silico pathogenicity (PP3, PM2). These entries have high REVEL scores (≥ 0.751), elevated CADD scores (≥ 25), and unanimous *likely_pathogenic* classification by AlphaMissense. These are the following entries:

Variant Notation	Protein Change	REVEL	CADD PHRED	AlphaMissense	gnomADe AF	ACMG Evidence
NM_002478.5:c.116A>G	p.Asp39Gly	0.751	31.0	likely_pathogenic	4.1×10^{-6}	PP3, PM2
NM_002478.5:c.248C>A	p.Ala83Glu	0.800	33.0	likely_pathogenic	2.1×10^{-6}	PP3, PM2
NM_002478.5:c.278G>A	p.Cys93Tyr	0.936	34.0	likely_pathogenic	*Absent*	PP3, PM2
NM_002478.5:c.327C>A	p.Asp109Glu	0.755	26.3	likely_pathogenic	2.1×10^{-6}	PP3, PM2
NM_002478.5:c.469A>G	p.Ile157Val	0.912	28.1	likely_pathogenic	1.2×10^{-5}	PP3, PM2
NM_002478.5:c.590C>T	p.Ser197Phe	0.777	29.7	likely_pathogenic	3.6×10^{-6}	PP3, PM2
NM_002478.5:c.611C>G	p.Ser204Cys	0.829	33.0	likely_pathogenic	1.5×10^{-5}	PP3, PM2

The remaining 16 variants in the dataset represented intermediate or ambiguous evidence categories. A total of 11 variants (20.4%) triggered PM2 alone due to their absence or extreme rarity in gnomAD ($AF < 0.000015$), yet fell into intermediate computational score ranges that precluded definitive PP3 or BP4 assignment. Finally, 5 variants (9.3%) exhibited conflicting predictor outputs or intermediate REVEL scores (0.561- 0.835) without consensus from AlphaMissense, categorizing them as ambiguous (No Specific Code). This computational evidence curation framework successfully resolved 90.7% of the *MYOD1* VUS dataset into specific ACMG evidence combinations, prioritizing the 7 high-scoring candidates for downstream functional validation in Rhabdomyosarcoma models.

3.2 Predictor Concordance and Distribution

Evaluating REVEL against CADD PHRED demonstrates a strong positive correlation among high-scoring damaging variants.

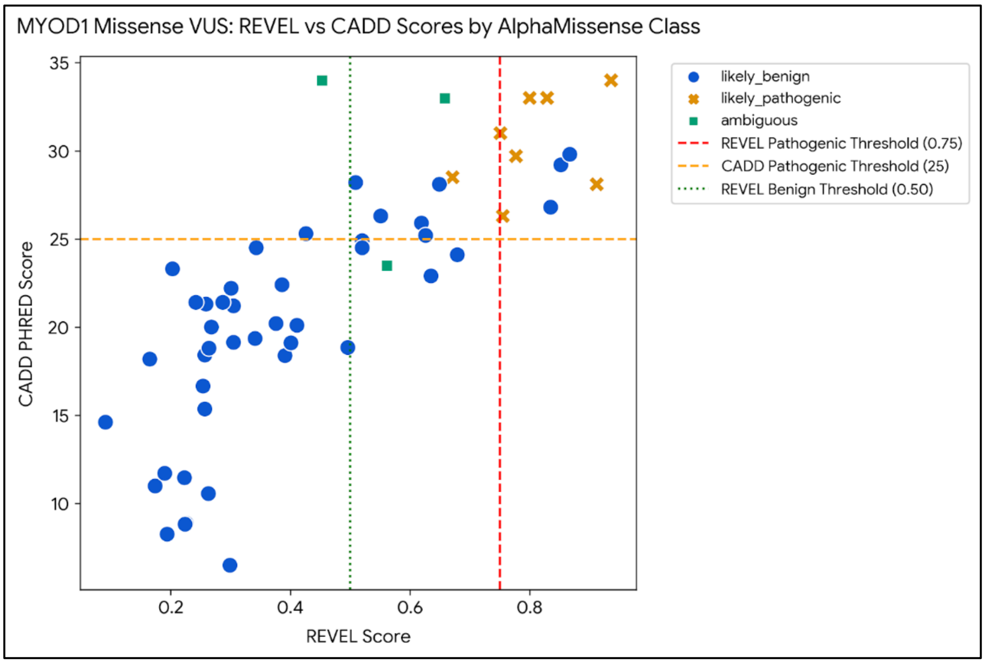


Figure 1: Scatter plot of REVEL vs. CADD PHRED scores for 54 *MYOD1* missense VUS categorized by AlphaMissense predictions. The upper-right quadrant isolates candidates exceeding both REVEL (≥ 0.75) and CADD (≥ 25) pathogenic thresholds.

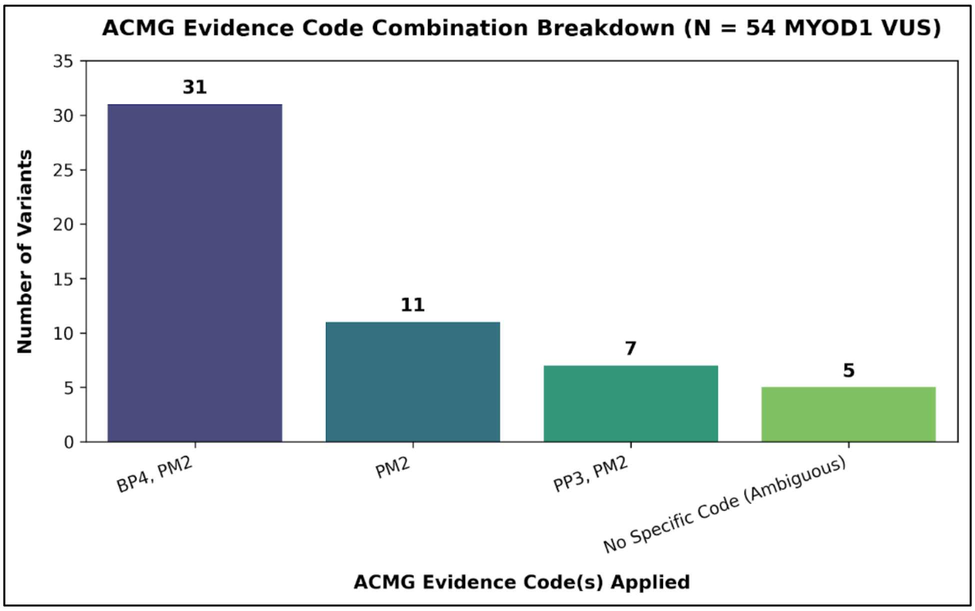


Figure 2: Distribution of ACMG evidence code combinations across the curated 54 *MYOD1* missense variants.

4. Discussion & Limitations

4.1 Biological Significance

This study successfully prioritised 7 missense variants in *MYOD1* that exhibit strong computational signals of damaging protein function. These mutations lie within or adjacent to critical structural domains of *MYOD1*, including the transactivation domain and the bHLH DNA-binding core. Conversely, over half of the ClinVar VUS entries (31 variants) met benign-leaning criteria (BP4), indicating that current clinical uncertainty for these specific variants may be driven by a lack of literature reports rather than actual functional disruption.

4.2 Limitations & Future Directions

Computational algorithms are evidence lines, not definitive proof. Studies indicate that ~11% of known benign variants can be falsely flagged as damaging by ensemble algorithms. In-silico evidence (PP3/BP4) alone cannot reclassify a VUS to Pathogenic or Benign in a clinical setting [6]. The 7 high-priority variants identified here require experimental testing, such as:

- Electrophoretic Mobility Shift Assays (EMSA) to test DNA binding capacity [8].
- Luciferase Reporter Assays in myogenic cell lines (e.g., C2C12 or RMS lines) to measure transactivation efficiency [8].
- Myogenic Differentiation Assays evaluating muscle-specific marker expression (e.g., Myogenin, MHC) [9].

References

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